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Intervention

Walking on prescription: The utility of a pedometer pack for increasing physical activity in primary care

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ABSTRACT

Objective: This study used a multi-method approach to examine the effectiveness of a pedometer-based intervention delivered by health professionals for increasing walking.
Methods: Pedometer packs were distributed to 374 patients who undertook a 12-week walking programme. Changes in walking were assessed at three months (using self-reported step-counts [$n = 139$] and questionnaire data [$n = 104$]) and at six months (using patient questionnaire data [$n = 112$]). Qualitative data were collected at both time points to identify underlying mechanisms.
Results: After 12 weeks patients had increased their step-counts by 4532 steps/day ($p < .001$) and at six months were achieving 2977 more steps/day (equivalent to around 30 min/day) than at baseline. Over half the patients reported achieving this target on at least 5 days/week. Qualitative data indicated that the pedometer pack was perceived to be most effective when patients were ready to change and when ongoing support was made available.
Conclusion: These findings support the use of pedometer-based interventions in primary care and suggest that the pedometer pack could lead to sustainable changes in walking. Further investigation, using a randomised controlled trial design, is warranted.
Practice implications: Provision of social support and accurate identification of patient readiness to change are important considerations in future implementation of the intervention.
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1. Introduction

The health benefits of regular physical activity have been consistently demonstrated across populations [1–3]. National and international guidelines recommend that adults accumulate at least 30 min of moderate intensity activity, on at least five days of the week, to promote and maintain health [3–5], and it is also likely that this level of energy expenditure will contribute to ongoing weight management [3]. However in Scotland, six out of ten men and seven out of ten women do not achieve this target [6].

Brisk walking has been described as 'the nearest activity to perfect exercise' [7] and is suggested as the mode of activity most likely to increase physical activity at a population level [8]. A recent systematic review [9] examined the effectiveness of interventions aimed at increasing walking and concluded that the strongest evidence exists for tailored interventions that are targeted at individuals most motivated to change, particularly involving strategies such as brief advice and the supported use of pedometers.

Pedometers are considered valid and reliable tools with which to measure physical activity [10,11]. A large number of studies have used pedometers to promote physical activity within university and community settings, and they have also been used among patients with diabetes [12], those with osteoarthritis [13], and within general practice [14].

Findings suggest that short-term effectiveness of pedometer programmes is enhanced when a goal-setting programme is incorporated [15–17], and, although there is no clear evidence, individual goals are thought to enhance motivation more than a standardised 10,000 steps/day target. Based on the assumption that 1000 steps approximates to 10 min of moderate activity [17], a realistic goal for inactive individuals is the gradual accumulation of an additional 3000 steps on most days of the week, leading to successful achievement of the minimum recommended levels of physical activity. Despite promising short-term changes in behaviour, there is a lack of evidence to support a long-term increase in physical activity following a pedometer intervention [15].

Primary care is a valuable and effective setting in which to promote physical activity [18–20]. Moreover, heath professionals (HPs) are well positioned to disseminate physical activity-related messages among large numbers of patients [21], who in the UK are

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entitled to free health care through the National Health Service. One of the key priorities in Scotland's physical activity strategy is that adults who come into contact with primary care should be offered an assessment of the health risks associated with their level of physical inactivity and provided with individualised help to increase their activity level by, for example, physical activity 'on prescription' [5].

Two UK-based studies have provided provisional evidence for the short-term utility of physical activity on prescription using pedometers and goal-setting programmes within primary care [22,23]; however, the effectiveness of such programmes over the longer term [15] and within different clinical settings has yet to be determined. In light of this limitation, Paths to Health, a Scottish programme that promotes walking, developed a resource for HPs to promote physical activity. The resource, or 'pedometer pack', contains a pedometer, step-count card, and supporting materials. Pilot data suggest that the pedometer pack is an accessible and resource-light tool and that it can help to initiate conversations about physical activity and lead to an increase in independent walking [23]. Given the value of the primary care setting for promoting physical activity, determining the potential applicability of the pedometer pack is an important line of research.

The purpose of this study is to further examine the utility of the pedometer pack in primary health care settings with patients who could benefit from increasing their physical activity, and add to the literature on the most effective methods for promoting walking. A mixed-methods approach will be used to explore implementation, underlying processes and conditions for optimal effectiveness. To address the limited literature on the longer term effectiveness of pedometer interventions the longer term, as well as short-term, effectiveness of the pedometer pack will be investigated.

2. Methods

2.1. Pedometer pack

The intervention involved two types of pedometer pack, a 'master pack' which was used by HPs and a patient pedometer pack (hereafter referred to as 'pedometer pack'). The master pack comprised of a briefcase-style box containing 10 basic pedometers (Silva Step Counter 56012; to the authors' knowledge, no reliability or validity data is available for this model), step-count record cards and supporting materials (e.g. leaflet on the benefits of walking, HP guidance, and promotional posters). Pedometers were encased within pillbox-style containers displaying a jovial warning that side effects of pedometer use include increased energy levels, stress reduction, and maintenance of a healthy weight (http://www.pathstohealth.org.uk/healthcarepedometerpack.asp). The pedometer pack included a pedometer, additional security clip, pedometer instructions card, step-count record card, and leaflet on the benefits of walking.

2.2. Pedometer pack distribution

HPs from 21 primary health care sites across Scotland volunteered to trial the use of pedometer packs, following a nation-wide advertisement. A range of HPs responded, including general practice (GP) nurses, health visitors, and physiotherapists. 168 master packs were distributed to the 21 sites which included GP clinics and physiotherapy, where patients received individual consultations, and weight management, mens' health and smoking cessation groups, where patients received health care in a group setting. A HP at each site co-ordinated the distribution of the pack and received briefing on its administration and patient selection. Briefing sessions were conducted by a member of the research team and included provision of generic advice on the minimum recommended levels of physical activity and identification of patients with conditions known to benefit from participation in regular physical activity. Over a six-month recruitment period, and based on advice given in briefing sessions, HPs issued pedometers to those patients who could benefit from raising their physical activity levels. Although it was not a formal aspect of the intervention, patients who received the pedometer pack in a group setting gained ongoing support from the HP and fellow group members. This support was not available for patients who received the pedometer pack in non-group settings.

2.3. Step-count targets

The step-count record card guided patients through an individualised walking programme consisting of four incremental step-goals, each of which patients aimed to complete within 1–3 weeks (see Table 1). The walking programme could therefore be completed in between 4 and 12 weeks. Following completion of stage 4, patients were advised that they should continue to achieve this target on at least 5 days of the week.

2.4. Data collection

The study was approved by the appropriate NHS research ethics committees. While administering the pedometer intervention, HPs invited patients to participate in the research and assisted in the transfer of data between patients and the research team. Data were collected at three time points: baseline, at the end of the walking programme (referred to hereafter as 'three months') and six months after the start of the programme (six months). Baseline step-count was calculated as the average of three days' step-count values; this has been suggested to provide a sufficient estimate of weekly physical activity in adults in a field setting [24].

At three months, patients handed in completed step-count record cards to HPs who then forwarded them to the research team. At this time, patients also completed postal questionnaires designed to elicit views on the effectiveness of the pedometer pack for promoting walking, which were then forwarded to the research team. Further, four focus groups including 6–8 patients and seven semi-structured telephone interviews with HPs were conducted. These participants were selected using a convenience sampling strategy [25], which was based on their availability to attend focus groups and interviews.

Follow-up data were collected at six months via questionnaires and interviews. Specifically, patients completed a postal questionnaire, which was then returned to the research team via HPs, containing 5-point Likert-type scales designed to determine the perceived ease of use of the pedometer pack and its effectiveness for promoting walking. In addition, qualitative data were collected from patients to explore possible longer term benefits of the pedometer scheme in greater depth. Reminder letters and telephone calls were used at all time-points to facilitate the return of data.

Table 1
Description of the 12 week walking programme.

Stage	Step-count target
Stage 1	Baseline + 1500 steps to be achieved on 3 days of the week
Stage 2	Baseline + 1500 steps to be achieved on 5 days of the week
Stage 3	Baseline + 3000 steps to be achieved on 3 days of the week
Stage 4	Baseline + 3000 steps to be achieved on 5 days of the week

2.5. Data analysis

Step-count data were analysed using SPSS v.14.0 (SPSS, Inc., Chicago, IL). Inspection (using histograms and Kolmogorov–Smirnov test) of baseline data ($n = 129$) suggested a non-normal distribution. As specified in Section 2.2, the pedometer packs were prescribed to individuals whose HPs considered would benefit from increasing their physical activity. An average step-count of >10,000 steps/day classifies a person as 'active' [26]; therefore, six patients who reported an average baseline step-count greater than 10,000 step/day were excluded from further analyses. In addition, removal of these six participants provided a dataset that was normal in distribution. Data were analysed using a one-way repeated measures ANOVA of difference scores between stages, with an alpha value of .05 or less set as the criterion of significance.

To analyse data collected in focus groups and telephone interviews, conversations were transcribed verbatim by the first author and subject to content analysis following the procedures outlined by Berg [27]. Quotations are presented in the results to illustrate the main issues that emerged.

3. Results

3.1. Response rate

In total, 374 (22%) out of a possible 1680 patient pedometer packs were distributed. Although this was below the expected distribution, time constraints associated with the length of the recruitment period and patient appointments in GP settings should be acknowledged. Baseline data were obtained from 34% (129; males = 33, females = 70, not completed = 26) of patients, six of whom were excluded due to step-counts >10,000 steps/day. Complete three-month step-count data were obtained from 82.9% (107) patients (see Table 2). Analyses were performed on these 107 patients only. At this time, 80.6% (104) of patients returned a completed questionnaire. Patients ranged in age from 16 to 84 years (*Mean* 57.3 years, *SD* 13.9).

3.2. Pedometer step-count data at three months (n = 107)

The descriptive step-count statistics for each stage of the three-month intervention are presented in Table 2 and the increase in step-counts over this time is shown in the box and whisker plot in Fig. 1. On average, patients exceeded their step-count target at each stage of the walking programme, and by stage four, they accumulated an average of 4532 steps/day more than at baseline. A one-way repeated measures ANOVA indicated that the differences in step-counts across the four stages of the programme were significant ($F_{(2.67,\ 296.21)} = 41.860$, $p < .001$, partial $\eta = .27$). There were significant increases in step-counts between baseline and stage one ($F_{(1,\ 111)} = 87.51$, $p < .001$, partial $\eta = .44$), between stages two and three ($F_{(1,\ 111)} = 60.48$, $p < .001$, partial $\eta = .35$), and between stages three and four ($F_{(1,\ 111)} = 9.21$, $p = .003$, partial $\eta = .08$).

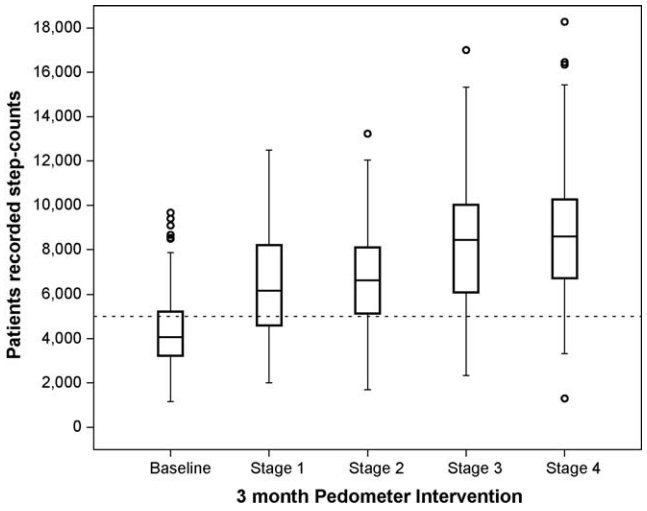


Fig. 1. Increase in average steps/day over the three month intervention; *y*-axis reference line at 5000 steps indicates the threshold below which patients can be classified as sedentary; outlier scores are represented by circles at the end of the distribution (30).

3.3. Patient questionnaire data at three months (n = 104)

Patients' views of the pedometer pack were generally very positive, with the pedometer and supporting materials being viewed as useful and easy to use. More specifically, 47.1% of patients indicated that the pedometer pack was 'very effective' in motivating them to become more physically active. The ease with which the recommended step-count targets could be attained was perceived by most patients (56.4%) to be 'moderately easy', although it should be noted that 9.9% found the targets to be 'not at all easy.' A large proportion (52.9%; 55) of patients reported that they were still using the pedometer at the end of the walking programme.

3.4. Interviews with HPs at three months

With regard to the design of the pedometer pack, all comments from HPs were positive. The pillbox style pedometer container was appreciated and viewed to 'get the message across well.' Moreover, the pedometer pack was seen as a useful tool for initiating conversations about physical activity, even where patients were not interested in participating in the walking programme.

Many HPs attributed the success of the pedometer pack to its potential to raise awareness of patients' physical activity levels. Typical feedback included:

Table 2
Step-counts for participants at each stage of the programme and changes in step-counts from baseline at each stage.

Stage and walking goal set	Number	*M*	*SD*	Minimum	Maximum
Baseline	123	4,422	1,848	1,000	9,667
Stage 1: +1,500 steps, 3 days a week	119	6,641	2,629	2,000	12,696
Stage 2: +1,500 steps, 5 days a week	118	6,877	2,358	505	13,231
Stage 3: +3,000 steps, 3 days a week	110	8,254	2,842	2,333	17,004
Stage 4: +3,000 steps, 5 days a week	107	8,906	3,151	1,300	18,279
Change in steps from Baseline to S1		2,224	1,712	−2,159	9,194
Change in steps from Baseline to S2		2,488	1,407	−3,850	6,354
Change in steps from Baseline to S3		3,851	2,164	−3,324	13,770
Change in steps from Baseline to S4		4,532	2,301	−445	12,256

'There's a lady who thinks that having a pedometer is a Godsend. She's lost two stone and done really, really well and she says it's because she could see at a glance what she was doing.' (HP from weight management clinic)

The effectiveness of the intervention was further attributed to the individualised approach to step-count goal setting, which was seen to 'give goals rather than ambiguous messages' and 'a less woolly' approach to increasing walking levels, than the commonly advocated 10,000 steps/day message.

Some HPs, while commending the pedometer pack for its effectiveness among those who were motivated to change behaviour, felt that the pack was of limited use among those who were not motivated to change:

'I think the pack is fantastic and I would definitely recommend it to anyone but it has to be with people who are ready to do it, that's the thing. There's absolutely no point in giving it to people who don't want to do it.' (HP from smoking cessation group)

This view was shared by practice nurses who had delivered the intervention within a GP setting. These HPs recognised the need for patient follow-up but felt that the time constraints of patient consultations limited the provision of such support. The format of group settings, however, allowed for the provision of ongoing support from both the HP and group members. HPs based in these settings viewed the intervention to be highly successful.

3.5. Patient focus groups at three months

Consistent with questionnaire responses, patients spoke positively about the ease of use of the pedometer pack. However, the perceived attainability of the step-count goals was equivocal, with some patients finding the incremental step-count goals 'difficult but not too difficult' and others finding progression between step-count stages 'too much of a jump'. Patients attributed a number of positive lifestyle changes to their involvement in the scheme, including walking to work and going on lunchtime and family walks. There was some indication that these changes were of a sustainable nature, as is reflected in the following comment:

'I see it [walking] very much as a new way of doing things and I won't go back now.' (patient from smoking cessation group).

Echoing feedback from HPs, raised awareness of activity levels and the provision of ongoing support motivated patients to increase their step-counts. Patients who had participated in the pedometer scheme within a group setting also identified the social support component from the HP and other members of the group as a key aspect of the scheme's success. One patient commented:

'I wouldn't have bothered with it if I'd been on my own; I'd have put it in a drawer. You've not got the same motivation on your own.' (patient from weight management group)

Extending this theme, those patients who received no peer support reported that they would have benefited from ongoing feedback:

'That's something I thought would have helped, you know, to have a dedicated person to encourage you . . . it was almost like it was up to you to go by yourself, whereas being in an environment where someone was continually monitoring progress might have worked better.' (patient from GP clinic)

3.6. Patient questionnaire data at six months (n = 112)

The return rate of postal questionnaires at six months was 86.8% (112; males = 34, females = 78). Responses indicated that 44% (49) of respondents were continuing to use the pedometer at the time of data collection. Of this group, 73.5% reported that they would continue to do so, while 20.4% and 6.1% respectively reported that they would 'probably' and 'possibly' continue to use the pedometer. These patients reported completing, on average, 7399 steps/day (*SD* 2894), 2977 steps/day more than baseline. The majority of patients (40.8%) reported achieving their average daily step-count on 5–6 days per week, while 38.8% reported achieving the step-counts on 2–4 days per week and 16.3% reported achieving the step-counts on all days of the week.

Of those patients who reported no longer using the pedometer, 96.7% reported that they were still walking for health benefits, most commonly on 2–4 days of the week (46%) and 5–6 days of the week (20.6%).

3.7. Patient qualitative data at six months

Qualitative data collection at six months proved to be problematic, and despite strong efforts to recruit patients, it was only possible to conduct three telephone interviews and one focus group with six patients. The data were, therefore, limited but provided some useful insights into reasons for continued and discontinued use of the pedometer. Those patients who were still using the pedometer at this time continued to view it as a motivational tool, while those who were no longer using the pedometer reported that the 'novelty' had worn off or that it had become an inconvenience. Importantly, however, most of these patients were continuing to walk for health benefits and no longer required the pedometer because they had learned how many steps were associated with their walking routes. Further supporting the potential of the intervention to lead to lifestyle and attitudinal changes, patients spoke of how walking had become an integral part of their daily lives. Typical comments included:

'I wouldn't have been interested in going for a walk maybe a couple of years ago; it's changed my lifestyle.' (patient from mens' health group)

4. Discussion and conclusion

4.1. Discussion

The pedometer pack investigated in this study was well received by both patients and HPs and was effective in producing significant increases in walking behaviour over the short- and longer term. These increases were sufficient to meet the minimum recommended levels of physical activity [4]. At three months, patients had increased their step-counts by 4484 steps to an average of 8906 steps/day, a shift from a sedentary lifestyle to being 'somewhat active' [26]. The average increase in step-counts following a pedometer intervention is 27% above baseline [15]. The current improvement of 101% is therefore a considerable finding, particularly in a primary care setting. A similar individualised 12-week step-count goal programme in a community setting produced a 47% increase in step-counts but included a physical activity consultation [28].

At six months, patients had maintained increased walking levels and were achieving 2977 more steps/day than at baseline (59% increase), equivalent to nearly 30 min of brisk walking [17]. While more than 50% of patients no longer used the pedometer at six months, the majority of these patients were still walking for health benefits at least twice a week. Data suggested that use of the pedometer pack had been instrumental in stimulating positive behaviour changes and that walking had become an integral part of patients' daily lives.

Highlighting the benefits of pedometers as tools that can provide immediate feedback [29,30], the potential for the

pedometer pack to raise awareness of physical activity levels was key to motivating patients to increase their walking levels. The design of the pedometer pack, reflecting the idea of walking as a prescription for better health, represented a further engaging element of the intervention and supports previous work endorsing physical activity recommendations presented on a prescription pad [31].

Consistent with a pedometer intervention among university staff [16], quantitative analysis indicated significant increases in step-counts on a week by week basis. Adding support to the use of short- and long-term goals rather than an overall target [32], qualitative data suggested that the individualised nature of step-count goals contributed to the intervention's effectiveness. Patients' views on the benefits of using the pedometer for feedback were consistent with literature indicating that performance enhancement through goal setting is gained only with immediate confirmation of the extent to which goals have been achieved [33,34]. As some patients found the pre-set targets too demanding, it is recommended that HPs are provided with the knowledge and skills to adopt a flexible approach in terms of programme length.

Feedback from HPs and patients illuminated the conditions for optimal effectiveness of the pedometer pack scheme and indicated that patients will reap the greatest benefits when they are ready to change behaviour. The importance of readiness to change has long been recognised, although evidence for whether stage-matched interventions along the lines of the transtheoretical model of behaviour change [35] are more successful than non stage-matched interventions is equivocal [36,37] and has not been confirmed for moderate lifestyle activities [38,39]. However, given that HPs' lack of self-efficacy with regard to motivating patients to successfully change behaviour has been acknowledged as a barrier for promotion of physical activity [40,41], such an approach has implications for the training of HPs in physical activity promotion. Individuals classified as in 'precontemplation' (inactive and no intention of changing behaviour [35]) should be provided with information on the benefits of physical activity which may in turn increase their intention to change; an intervention such as the pedometer pack could be implemented among those who have the intention to increase physical activity through walking.

Qualitative data also indicated that the intervention was better received when delivered in a group setting where ongoing support is available. Incorporating ongoing support from peers and HPs would be an important component of the development of the pedometer pack scheme. Social support is positively related to physical activity level [42–45], and interactive physical activity counselling and follow-up encouragement has elicited greater long-term maintenance of activity change than a leaflet or brief advice from a GP, particularly for individuals reporting low levels of social support [46,47]. In this study, ongoing support was problematic in GP settings due to time constraints associated with patient appointments. Lack of time has been highlighted by Scottish primary care staff as a barrier to promoting physical activity [48], and it has been conceded that even telephone support is not feasible in many primary care settings [14]. Yet, the likelihood of practice nurses prescribing physical activity has been shown to double when routine 10 min patient consultations are extended by between 1.5 and 2 min [49]. This must be taken into account if promotion of physical activity is to become firmly embedded within the health promotion framework.

One of the strengths of this study is its mixed-methods approach. The inclusion of qualitative data has provided an insight into the impact of the pedometer pack intervention and uncovered some of the processes that explain how it works and how it may be developed. Nevertheless, the study is subject to some limitations which should be considered when interpreting findings. Firstly, despite considerable efforts to retrieve data, the initial return rate of baseline data was relatively low and the possibility that only the most motivated and successful individuals returned step-count cards cannot be discounted. The effect of the pedometer pack for increasing walking levels is, therefore, not known for those patients who did not return data. However, the sizeable increase achieved by a relatively large number of individuals is still an important public health finding. Secondly, the pre-post design which did not include a control group prevents definitive conclusions from being drawn about the effects of the pedometer pack. This study might, however, be seen as an important preliminary phase of a larger evaluation process which has built on pilot findings and identified the active components of the intervention and the underlying mechanisms by which they influence outcomes. Based on such a phased approach to evaluation, a definitive, randomised, controlled trial and long-term implementation would represent subsequent phases following further development of the pedometer pack scheme [50].

4.2. Conclusion

These findings indicate that the pedometer pack can lead to longer term increases in walking behaviour, although caution must be exercised when drawing this conclusion as the research design did not include a control group. The results suggest that the intervention is most effective when patients are ready to change their physical activity behaviour and when it is delivered in a group setting or when ongoing support is made available. This is currently difficult in GP settings.

4.3. Practice implications

The pedometer intervention investigated in this study shows promising results as a means of increasing physical activity levels within primary care settings. Future implementation of the pedometer pack scheme should take into account patient readiness to change and the provision of ongoing support, with use in a group setting where feasible. HPs require specialised training and sufficient time in order to perform these tasks effectively.

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